

Intelligent, Personalized Scientific Assistant via Large Language Models for Solid-State Battery Research

Published as part of ACS Materials Letters *special issue* "Emerging Investigators in Materials Science".

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Cite This: *ACS Materials Lett.* 2025, 7, 1807–1816



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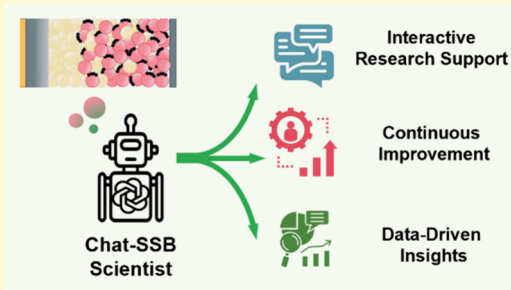


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Supporting Information

ABSTRACT: In response to the rapid advancements and heightened competition within solid-state battery research, the sheer volume of publications presents a significant challenge for researchers seeking comprehensive insights. This paper introduces ChatSSB, an advanced research assistant designed to bolster scientific inquiry within this dynamic field. Leveraging the Retrieval-Augmented Generation (RAG) framework, ChatSSB excels in extracting precise information from the latest research publications through an intuitive Q&A interface. Beyond its foundational capabilities, ChatSSB boasts a customizable expert knowledge database, continuously updated through a dynamic feedback mechanism. This ensures researchers have access to cutting-edge and reliable information, overcoming the limitations of outdated or incomplete literature. Furthermore, the integration of multiagent collaboration and embedded tools within RAG facilitates robust quantitative analysis, enabling efficient data collection, visualization, and interpretation. Collectively, these features empower ChatSSB to deliver precise, actionable insights, significantly accelerating innovation in solid-state battery technology and propelling it toward the next frontier of materials science.



With the global surge in clean energy demand, renewable energy technologies have emerged as pivotal solutions in mitigating climate change and addressing energy crises.¹ Central to these solutions are batteries, particularly lithium-ion batteries, which are known for their high energy density and longevity. These batteries have become ubiquitous across electric vehicles, portable electronics, and energy storage systems.^{2,3} However, the evolution of conventional lithium-ion batteries is inhibited by inherent safety and energy density limitations, necessitating the pursuit of more advanced and secure energy storage mechanisms.^{4,5} All-Solid-State Batteries (ASSBs), employing solid electrolytes, are at the forefront of this next-generation of high-energy-density batteries. Incorporating solid electrolytes presents significant improvements in safety, curbs lithium dendrite formation, and ensures stability at high energy densities and voltages.^{6–10} Despite their potential, the path to commercializing ASSBs is fraught with challenges,¹¹ including the inadequacy of current solid electrolytes^{12–14} and persistent interface issues between electrolytes and electrodes, which hinder cycle performance and safety.^{15–17} Additionally, the nascent manufacturing techniques required

for ASSB demand innovation to align with cost efficiencies seen in traditional lithium-ion battery production.¹⁸

In response, nations globally such as the United States,¹⁹ China,²⁰ Japan, South Korea, and European nations²¹ are introducing policies and initiatives aimed at advancing ASSB technology, an endeavor mirrored by the exponential increase in related academic publications. Yet, this proliferation of literature and data (i.e., novel material development to process optimization²²) overwhelms conventional research methodologies, creating an urgent need for tools that efficiently distill and synthesize vast information to actionable insights.²³ Enter artificial intelligence, specifically Large Language Models (LLMs), which are revolutionizing the landscape of scientific research by transcending traditional data handling limitations.^{24–27} These models, renowned for their ability to process

Received: December 30, 2024

Revised: April 6, 2025

Accepted: April 7, 2025

Published: April 9, 2025

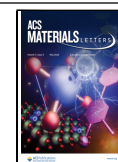




Figure 1. Overview of ASSB challenges and the ChatSSB framework for intelligent research support.

complex linguistic structures and generate high-quality content, surpass earlier natural language processing (NLP) techniques. However, even state-of-the-art foundational models still fall short in handling specialized domain knowledge.^{28–32} Retrieval-Augmented Generation (RAG) models further enhance LLM functionality by incorporating retrieval mechanisms that consistently source the latest pertinent data, minimizing typical AI deficiencies like hallucinations^{33–36} and avoiding catastrophic forgetting issues which are common in fine-tuned LLMs.^{37–39}

While RAG models represent a significant advancement by achieving high accuracy without additional training costs,^{40,41} they remain in the early stages of development, with key limitations.^{42,43} One major challenge is their reliance on the updates of the literature database, which hampers their ability to rapidly adapt to and incorporate the latest scientific advancements. Furthermore, traditional models struggle to access insights from expert discussions at academic workshops and conferences, resulting in significant knowledge gaps that impede timely and relevant information dissemination. Moreover, although LLMs have demonstrated remarkable progress in NLP, they often fall short in executing complex tasks, such as data collection and visualization, particularly in specialized fields where precision is critical. These shortcomings underscore the urgent need for an advanced tool that can seamlessly integrate comprehensive data acquisition and expert insights.

In this context, we designed ChatSSB, an intelligent and customizable research assistant, to address these challenges by harnessing both LLMs and RAG technology, thereby streamlining and elevating the materials discovery process for ASSBs. As illustrated in Figure 1, ChatSSB leverages a dual-source

approach, drawing from current academic literature and a continuously updated expert knowledge database, ensuring access to the most relevant and reliable information. This integration, coupled with a feedback system that enhances accuracy over time, positions ChatSSB as a crucial tool for researchers. By employing multiagent collaboration and embedded tools for comprehensive data collection, visualization, and analysis, ChatSSB delivers precise insights that propel advancements in ASSB technology, providing a competitive edge in this rapidly evolving domain.

ChatSSB is strategically architected around two integral databases and a dynamic feedback module designed to meet the evolving needs of researchers in the ASSB domain. The literature database forms the cornerstone of ChatSSB's knowledge base, featuring a comprehensive and up-to-date collection of relevant research findings. This ensures that ChatSSB can provide users with the latest advancements, technical developments, and experimental data in ASSB research. Complementing this is the expert knowledge database, which curates customized information sourced from domain experts, incorporating real-time feedback from users during interactions. This dual system enhances the accuracy of ChatSSB's responses, enabling it to offer insightful and precise answers tailored to specific inquiries. The feedback module further amplifies this capability by allowing the system to continuously evolve based on user interactions. After each engagement, ChatSSB optimizes its performance through an iterative learning process, improving both the accuracy and the consistency of its outputs over time.

When a user poses a question, as shown in Figure 2a, ChatSSB first retrieves pertinent literature from its ASSB

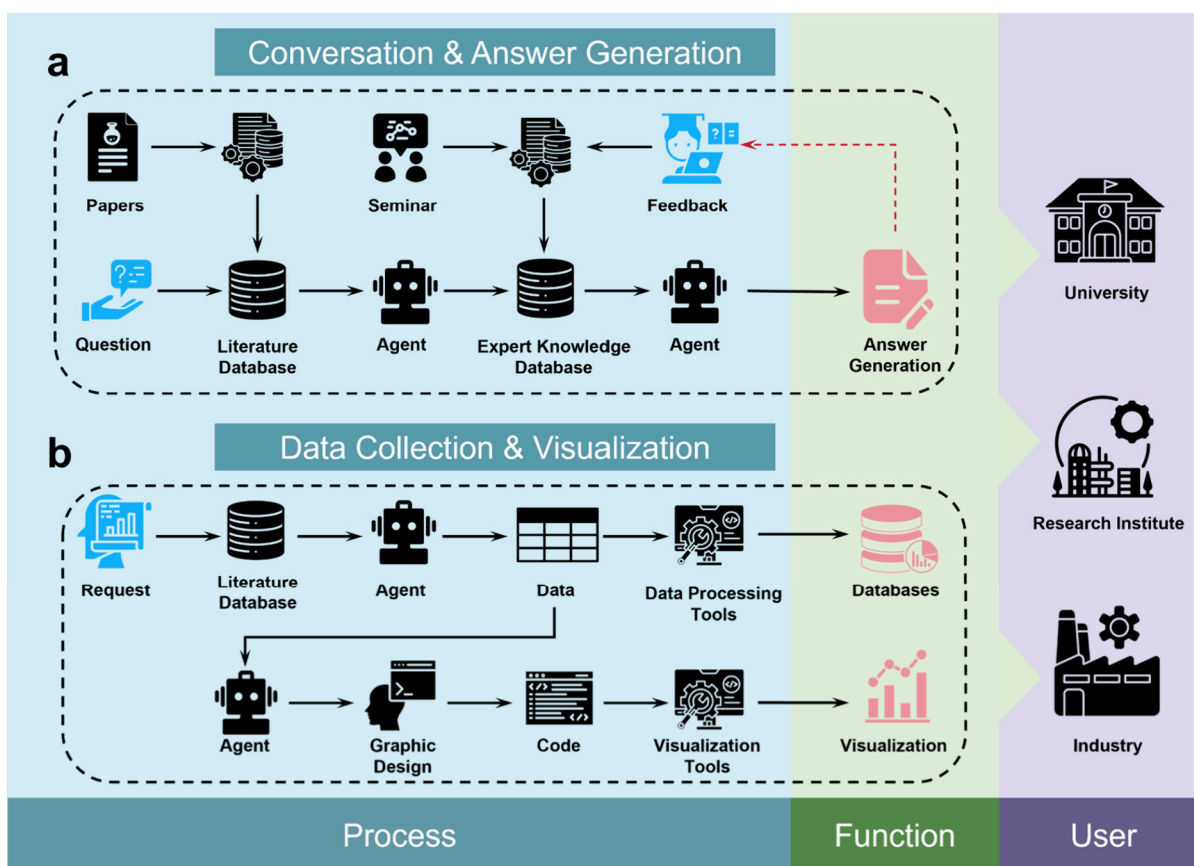


Figure 2. ChatSSB workflow: reducing hallucinations with RAG and streamlining research data collection and visualization. (a) RAG architecture for reducing hallucinations in ChatSSB. (b) Data collection and visualization workflow in ChatSSB.

database, extracting critical technical details and experimental results. This information is then cross-referenced with insights from the expert knowledge database, synthesizing expert opinions and user feedback to generate a comprehensive, accurate, and informative response. This dual retrieval and feedback mechanism serves to minimize hallucinations, ensuring that researchers receive reliable, high-quality information vital for their work.

In addition to addressing hallucinations, ChatSSB significantly enhances the efficiency of scientific research through its RAG functionality and multiagent collaboration, augmented by embedded research tools. These features enable users to systematically collect, organize, analyze, and visualize research data from diverse and collective resources. As illustrated in Figure 2b, when a user requests specific data collection, ChatSSB employs its advanced RAG architecture to efficiently retrieve relevant research information from the literature database. Utilizing the capabilities of large language models, the system extracts key insights from an extensive body of literature, ensuring comprehensiveness and accuracy in the retrieved data. Once collected, the data can be utilized to generate tables or establish new data sets, providing valuable references for ongoing research. In terms of data visualization, ChatSSB demonstrates robust capabilities tailored to user needs. Upon receiving a request for tables or charts, the primary agent first gathers and processes the relevant data. This information is then relayed to a secondary agent, responsible for designing the visual representation based on the user's specific demands—selecting appropriate chart types and presentation styles. Ultimately, ChatSSB generates the

corresponding code that is subsequently executed through the system interface to accomplish the data visualization, delivering a clear and intuitive chart to the user. This automated process not only conserves significant time and effort for researchers but also ensures accurate and efficient data collection and visualization.

We present a detailed demonstration of how ChatSSB assists researchers in the evolving field of ASSB. This process is visually articulated in Figure 3, which highlights ChatSSB's capabilities in addressing key research challenges and providing effective solutions. To illustrate its exceptional question-answering abilities and knowledge integration functionalities, we have devised a series of representative Q&A examples. For instance, when confronted with fundamental inquiries about ASSBs, researchers may pose questions such as “What are the primary research directions in solid-state batteries?” or “What types of high-performance solid electrolytes used in solid-state batteries?” In these scenarios, ChatSSB can swiftly and accurately generate comprehensive responses, covering a wide spectrum from fundamental concepts to the cutting-edge research in the field. Its robust database allows for high-quality responses, significantly reducing the time researchers spend searching for essential information and allowing them to focus on innovation.

To demonstrate ChatSSB's capabilities further, we explored the domain of specific advanced solid-state electrolyte (SSE) materials, with a particular focus on garnet-type SSE. Garnet-type SSEs, such as $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (LLZO),⁴⁴ are among the most extensively studied candidates for safe, high-energy ASSBs due to their high ionic conductivity and remarkable

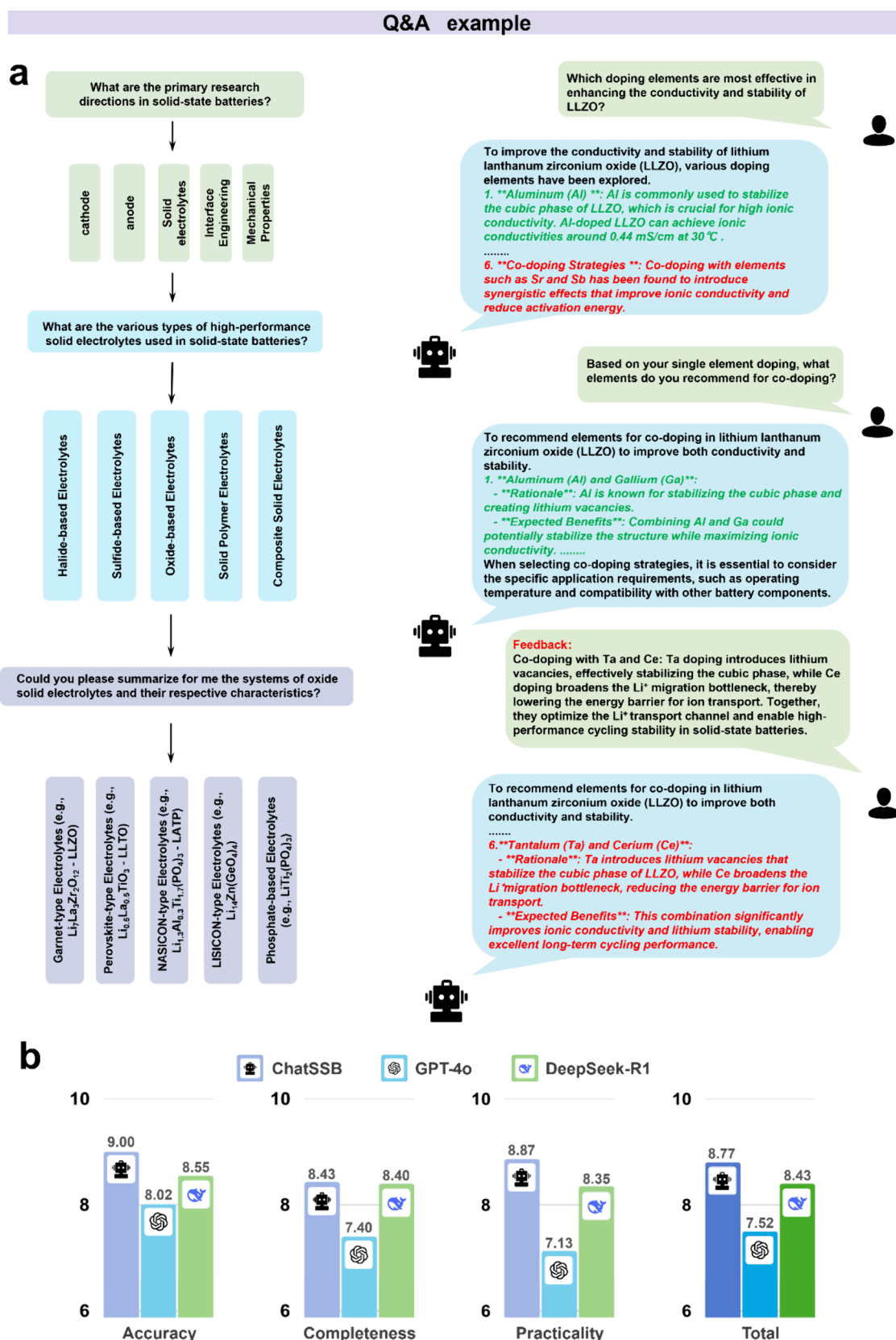


Figure 3. Demonstration of ChatSSB's performance in ASSB research. (a) Demonstration of ChatSSB's advanced Q&A capabilities and expert feedback integration in ASSB research. (b) Comparison of model performance based on combined ratings for accuracy, information completeness, and practicality.

chemical stability in contact with lithium metal.⁴⁵ Through a multiturn conversation, as shown in Figure 3a, we illustrate

how ChatSSB effectively engages with users. A researcher might initiate the dialogue by asking, "Which doping elements

Data Extraction & Visualization

a

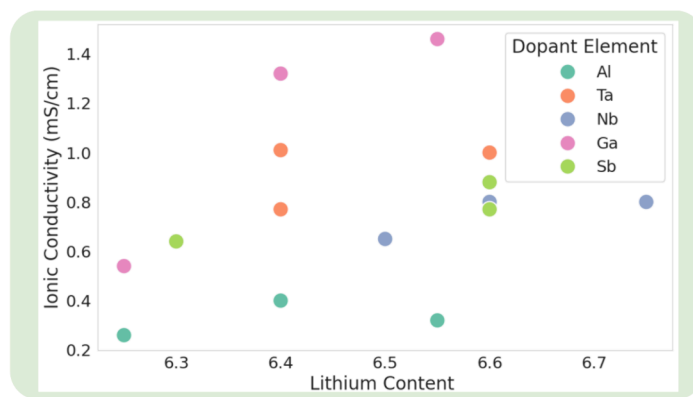
State the chemical formulas of different garnet solid electrolytes other than $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$, their specific synthesis methods including precursors and synthesis conditions, and specific values of ionic conductivity, and present this information in a table.



No.	Chemical Formula	Synthesis Method	Precursors	Synthesis Conditions	Ionic Conductivity (mS/cm)
1	$\text{Li}_{6.5}\text{La}_3\text{Zr}_{1.5}\text{Ta}_{0.5}\text{O}_{12}$	Hot pressing	Li_2CO_3 , La_2O_3 , ZrO_2 , Ta_2O_5	Pressed at 30 MPa, sintered at 1230°C for 6 hours	0.87
2	$\text{Li}_{6.10}\text{Ga}_{0.30}\text{La}_3\text{Zr}_2\text{O}_{12}$	Solid-state reaction	Li_2CO_3 , Ga_2O_3 , La_2O_3 , ZrO_2	Calcined at 1100°C for 12 hours	1.12
3	$\text{Li}_{6.4}\text{La}_3\text{Zr}_{1.4}\text{Ta}_{0.6}\text{O}_{12}$	Field-assisted sintering	Li_2CO_3 , La_2O_3 , ZrO_2 , Ta_2O_5	Sintered at 1200°C under 50 MPa pressure	1.01
4	$\text{Li}_7\text{La}_{2.75}\text{Ca}_{0.25}\text{Zr}_{1.75}\text{Nb}_{0.25}\text{O}_{12}$	Solid-state reaction	Li_2CO_3 , CaCO_3 , La_2O_3 , ZrO_2 , Nb_2O_5	Fired at 1150°C for 12 hours	0.22
5	$\text{Li}_{6.55}\text{Ga}_{0.15}\text{La}_3\text{Zr}_2\text{O}_{12}$	Solid-state reaction	Li_2CO_3 , Ga_2O_3 , La_2O_3 , ZrO_2	Calcined at 1200°C for 10 hours	2.06

b

Please collect information on ionic conductivity for LLZO solid electrolytes in the form of a table, then draw a figure showing this information. Try to show the influence of lithium proportion and doping elements on ionic conductivity.



c

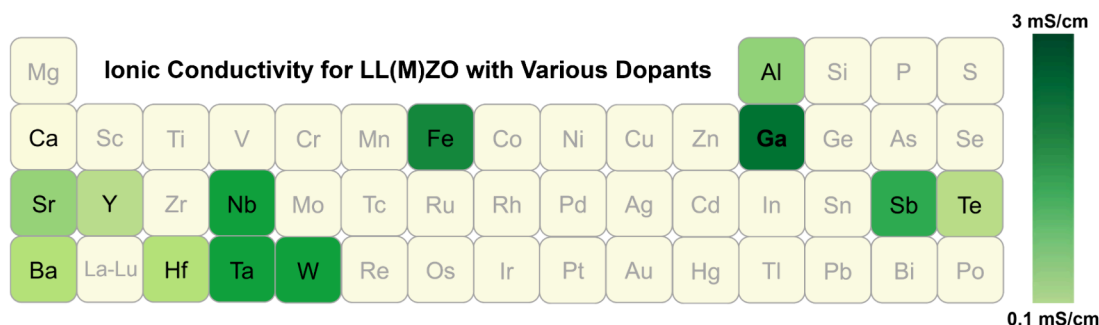


Figure 4. Streamlined data visualization with ChatSSB: Insights into ionic conductivity trends for LLZO solid electrolytes. (a) LLZO synthesis information table was generated by ChatSSB. (b) Visualization of the impact of lithium proportion and doping elements on LLZO ionic conductivity. (c) Comparison of maximum ionic conductivity achieved by LLZO doping strategies extracted from ChatSSB from the database.

are most effective in enhancing the conductivity and stability of LLZO?" In response, ChatSSB seamlessly integrates relevant research findings, providing a detailed explanation of the effects of single-element doping while also introducing the concept of codoping strategies. Building upon this interaction, the researcher may inquire, "Based on your single-element doping, what elements do you recommend for codoping?" In this case, ChatSSB retrieves additional contextual information, delivering a more comprehensive answer regarding codoping approaches. Following this exchange, researchers can provide feedback on ChatSSB's responses,⁴⁶ which the system systematically collects and incorporates into its knowledge base. Once the researcher provides feedback, the system processes it by generating a concise summary and storing it in a structured format. The system systematically collects and incorporates into its knowledge base, ensuring that future responses better reflect user expectations and accumulated expert insights transforming it into a self-evolving, personalized research assistant. In summary, ChatSSB significantly enhances the efficiency and quality of research for scientists in the ASSB field through its advanced Q&A capabilities and multiturn conversation support. By intelligently integrating and updating data, ChatSSB offers personalized assistance tailored to the customized needs of researchers, ensuring they maintain a competitive edge in the rapidly advancing field of ASSB research.

To comprehensively assess the performance of ChatSSB in the field of solid-state batteries, we compared it against its base model, GPT-4O, and one of the most advanced models currently available, DeepSeek-R1. To ensure a thorough and scientific evaluation, we designed a test set consisting of 30 specific questions related to solid-state batteries and electrolytes. These questions covered core aspects of solid-state battery technology, such as material performance, structural design, and electrolyte properties, aiming to evaluate the models' ability to handle complex technical problems. In the evaluation process, we focused on three key criteria: accuracy, information completeness, and practicality, which reflect the models' mastery of knowledge as well as their effectiveness in real-world applications. Each criterion was rated on a scale from 1 to 10, and a final score was calculated by averaging these scores. We used GPT-o1 as the Judge model to evaluate the performance of all three models.

Considering the combined ratings for accuracy, information completeness, and practicality, ChatSSB emerged as the clear winner with an overall score of 8.77/10, outperforming DeepSeek-R1 (with a score of 8.43/10) and GPT-4O (with a score of 7.52/10). ChatSSB demonstrated clear advantages in theoretical accuracy, breadth and depth of information, and the ability to integrate these insights into practical, real-world scenarios, making it the most outstanding model in this evaluation. It is worth noting that ChatSSB obviously outperforms DeepSeek-R1, although its base model, GPT-4O, is inferior to that of DeepSeek-R1 in our tasks. This is attributed to the efficacy of the designed RAG mechanisms and embedded tools, which have effectively enhanced the capability of ChatSSB for addressing the challenges in specific domains. Through this evaluation, we have not only validated ChatSSB's depth and comprehensiveness in theoretical analysis but also highlighted its unique advantage in providing practical solutions for engineering applications. In contrast, although GPT-4O and DeepSeek-R1 performed well in several areas, they fell short in depth and applicability. Therefore, ChatSSB

undoubtedly claims the leading position in this evaluation, making it the most exceptional tool for answering questions in the field of solid-state batteries.

In the field of ASSB research, experimental investigations are essential for understanding and optimizing material performance, encompassing factors such as synthesis parameters and electrochemical characteristics. The development of high-performance SSEs is particularly influenced by the choice of synthesis methods and parameters, directly affecting the properties of the target materials. Therefore, researchers often invest significant time and effort in reviewing the literature, designing experiments, and aggregating experimental data to extract valuable insights.

For instance, when researchers seek to gather synthesis parameters for garnet-type SSEs such as $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (LLZO), the traditional paradigm often necessitates sifting through numerous publications and reading through lengthy texts. This process can be both time-consuming and labor-intensive. In contrast, with ChatSSB, researchers can simply ask "State the chemical formulas of different garnet solid electrolytes other than $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$, their specific synthesis methods including precursors and synthesis conditions, and specific values of ionic conductivity, and present this information in a table." In response, the ChatSSB quickly compiles a comprehensive table of key information, which includes chemical formulas, synthesis methods, precursors, synthesis parameters, and ionic conductivity values. As shown in Figure 4a, this features seamless comparison of various garnet SSE compositions, enabling researchers to select the most appropriate experimental strategy.

Furthermore, ChatSSB can also enhance research efficiency through its powerful data visualization features, which allow researchers to intuitively interpret results and identify key trends. For example, when prompted with "Please collect information on ionic conductivity for LLZO solid electrolytes in the form of a table, then draw a figure showing this information. Try to show the influence of lithium proportion and doping elements on ionic conductivity." ChatSSB adeptly manages the entire workflow from data collection to visualization. First, ChatSSB systematically gathers pertinent data from the scientific literature, organizing essential information into a clear and detailed table for easy review and verification. Building on this data collection, ChatSSB generates Python code to create a scatter plot that visually represents the relationships among these variables. Using packages such as Matplotlib or Seaborn, ChatSSB designs a plot specifically tailored to highlight significant patterns. The resulting scatter plot, as illustrated in Figure 4b, effectively demonstrates how lithium content and doping elements influence ionic conductivity, revealing important trends, such as the influence of lithium content on ionic conductivity, as well as the significant improvements achieved through doping with elements like Al and Ga. Additionally, Figure 4c complements this analysis by visualizing the ionic conductivity values of LLZO associated with different dopants. ChatSSB extracts this information from its expert knowledge database, pinpointing the top-performing doping strategies and their corresponding ionic conductivity values. Notably, the highest ionic conductivity recorded in the database—2.06 mS/cm—was observed in Ga-doped LLZO.⁴⁷ This visualization offers researchers a clear reference for selecting the most effective doping elements to optimize LLZO's performance.

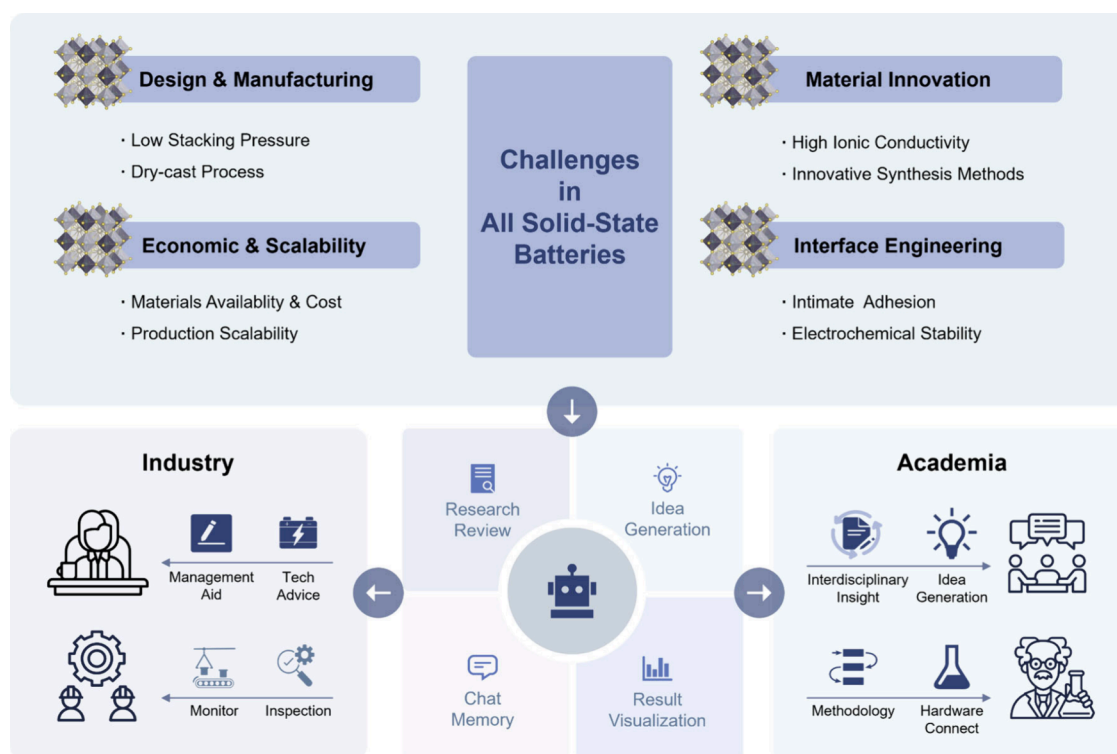


Figure 5. Future prospects of ChatSSB in academia and industry.

By streamlining data collection, analysis, and visualization into an integrated process, Chat SSB eliminates the need for labor-intensive manual operations such as searching through papers and extracting data. This allows researchers to focus on interpreting the results and designing experiments rather than on being encumbered by preparatory work. By seamlessly merging these functions, ChatSSB empowers researchers to efficiently explore essential relationships and generate valuable insights, thereby significantly accelerating the progress in the field of ASSB.

ChatSSB leverages its scalable RAG architecture to integrate diverse data sources with flexibility, positioning itself as a versatile tool capable of addressing a wider range of challenges across disciplines. This adaptability enhances ChatSSB's functionality, empowering it to deliver targeted solutions for both academic research and industrial innovation.

Beyond SSB research, ChatSSB's modular framework can be adapted to diverse fields.⁴⁸ In energy storage, expanding its knowledge base to materials such as ceramics or organic materials—coupled with agents analyzing composition—structure—performance relationships—could accelerate battery materials and electrolyte discovery. At the device level, ChatSSB could integrate system-wide data (e.g., electrode—electrolyte interfaces, thermal management, and aging mechanisms) to optimize full-cell configurations, predict cycle life under varied operating conditions, or balance trade-offs between energy density and safety. For biomaterials, integrating biocompatibility and degradation data would enable ChatSSB to optimize medical implants or drug delivery systems by correlating synthesis parameters and microstructures (e.g., porosity, cross-linking) with biological outcomes. Similarly, in advanced manufacturing, agents trained on processing structure—property data sets could guide additive manufacturing protocols for alloys or composites.

While this work focuses on SSBs, ChatSSB's architecture—combining customizable databases, specialized agents, and dynamic feedback—is universally applicable to research fields grappling with data fragmentation and complexity. By tailoring its knowledge base and analytical tools, ChatSSB can empower researchers in any discipline to extract actionable insights, bridge domain-specific knowledge gaps, and accelerate discovery. This framework represents a paradigm shift toward AI-augmented, data-driven science with transformative potential across academia and industry.

Moreover, ChatSSB's ability to autonomously connect with experimental hardware paves the way for closed-loop workflows, where real-time feedback from experiment informs rapid adjustments. This seamless integration of data processing with experimentation accelerates material synthesis and refinement,^{49,50} significantly enhancing research efficiency. In industrial applications, ChatSSB holds promise in overcoming challenges related to economic feasibility and scalability. By incorporating real-time feedback from production lines into its model, ChatSSB can compare these data against knowledge from its databases to identify deviations and recommend optimized solutions. Mirroring its conversational mechanism, ChatSSB could also establish a task-specific production line feedback database using past production outcomes to continuously refine its recommendations. This iterative approach allows ChatSSB to streamline manufacturing workflows by providing task-specific, data-driven guidance for immediate and precise adjustments.

As shown in Figure 5, ChatSSB has the potential to cultivate a novel ecology of research by integrating diverse data sets, enabling real-time optimization, and fostering cross-disciplinary collaboration. The transition from isolated data silos to a dynamic ecosystem of open sharing and collaborative innovation holds great promise for accelerating advancement

in various scientific and industrial fields, ultimately driving significant progress in the material science landscape.

In conclusion, this study presents ChatSSB, an intelligent and personalized scientific assistant specifically designed for ASSB research. By leveraging the RAG framework, ChatSSB effectively mitigates hallucinations while establishing itself as a reliable vertical-domain expert, proficiently extracting and synthesizing information from academic databases and expert feedback repositories. The integration of advanced prompting techniques, multiagent collaboration, and embedded analytical tools significantly enhances ChatSSB's functionality, enabling capabilities such as data visualization, precise knowledge retrieval, and support for experimental design. Through the incorporation of a unique expert knowledge base, ChatSSB enhances both the timeliness and personalization of its assistance, seeking a balance between static literature data and dynamic expert insights. Equipped with these advanced techniques, ChatSSB demonstrates the potential to foster meaningful advancements in SSB research. Case studies, such as its application in LLZO electrolyte research, highlight ChatSSB's ability to accelerate experimental planning and optimize decision-making processes. Moreover, ChatSSB's capabilities extend beyond ASSB research, laying the groundwork of transformative changes in data-driven research across various scientific fields. By integrating real-time experimental feedback and supporting laboratory automation, ChatSSB's utility can be further enhanced, promoting a more efficient research environment. In summary, ChatSSB enhances the research efficiency, accuracy, and accessibility within SSB studies. By combination of advanced artificial intelligence techniques with precise domain-specific customization, it represents a significant leap toward intelligent, automated scientific exploration. This positions ChatSSB as a foundational tool for future innovations in both academic and industrial applications, paving the way for breakthroughs that can dramatically influence the trajectory of materials science and energy storage technologies.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsmaterialslett.4c02674>.

Method, design of feedback module, full sample Q&A conversations, LLZO data visualization, and model evaluation section (PDF)

Model evaluation questions and answers ratings (PDF)

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<https://pubs.acs.org/doi/10.1021/acsmaterialslett.4c02674>

Notes

The authors declare no competing financial interest.

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